



**Figure 4.33** ♦ Steps in adding an outside-AS destination in a router's forwarding table

flexibility in deciding which destinations it advertises to its neighboring ASs. This is a *policy* decision, typically depending more on economic issues than on technical issues.

Recall from Section 1.5 that the Internet consists of a hierarchy of interconnected ISPs. So what is the relationship between ISPs and ASs? You might think that the routers in an ISP, and the links that interconnect them, constitute a single AS. Although this is often the case, many ISPs partition their network into multiple ASs. For example, some tier-1 ISPs use one AS for their entire network; others break up their ISP into tens of interconnected ASs.

In summary, the problems of scale and administrative authority are solved by defining autonomous systems. Within an AS, all routers run the same intra-AS routing protocol. Among themselves, the ASs run the same inter-AS routing protocol. The problem of scale is solved because an intra-AS router need only know about routers within its AS. The problem of administrative authority is solved since an organization can run whatever intra-AS routing protocol it chooses; however, each pair of connected ASs needs to run the same inter-AS routing protocol to exchange reachability information.

In the following section, we'll examine two intra-AS routing protocols (RIP and OSPF) and the inter-AS routing protocol (BGP) that are used in today's Internet. These case studies will nicely round out our study of hierarchical routing.

## 4.6 Routing in the Internet

Having studied Internet addressing and the IP protocol, we now turn our attention to the Internet's routing protocols; their job is to determine the path taken by a datagram between source and destination. We'll see that the Internet's routing protocols embody many of the principles we learned earlier in this chapter. The link-state and distance-vector approaches studied in Sections 4.5.1 and 4.5.2 and the notion of an autonomous system considered in Section 4.5.3 are all central to how routing is done in today's Internet.

Recall from Section 4.5.3 that an autonomous system (AS) is a collection of routers under the same administrative and technical control, and that all run the same routing protocol among themselves. Each AS, in turn, typically contains multiple subnets (where we use the term subnet in the precise, addressing sense in Section 4.4.2).

### 4.6.1 Intra-AS Routing in the Internet: RIP

An intra-AS routing protocol is used to determine how routing is performed within an autonomous system (AS). Intra-AS routing protocols are also known as **interior gateway protocols**. Historically, two routing protocols have been used extensively for routing within an autonomous system in the Internet: the **Routing Information Protocol (RIP)** and **Open Shortest Path First (OSPF)**. A routing protocol closely related to OSPF is the **IS-IS** protocol [RFC 1142, Perlman 1999]. We first discuss RIP and then consider OSPF.

RIP was one of the earliest intra-AS Internet routing protocols and is still in widespread use today. It traces its origins and its name to the Xerox Network Systems (XNS) architecture. The widespread deployment of RIP was due in great part to its inclusion in 1982 in the Berkeley Software Distribution (BSD) version of UNIX supporting TCP/IP. RIP version 1 is defined in [RFC 1058], with a backward-compatible version 2 defined in [RFC 2453].

RIP is a distance-vector protocol that operates in a manner very close to the idealized DV protocol we examined in Section 4.5.2. The version of RIP specified in RFC 1058 uses hop count as a cost metric; that is, each link has a cost of 1. In the DV algorithm in Section 4.5.2, for simplicity, costs were defined between pairs of routers. In RIP (and also in OSPF), costs are actually from source router to a destination subnet. RIP uses the term *hop*, which is the number of subnets traversed along the shortest path from source router to destination subnet, including the destination subnet. Figure 4.34 illustrates an AS with six leaf subnets. The table in the figure indicates the number of hops from the source A to each of the leaf subnets.

The maximum cost of a path is limited to 15, thus limiting the use of RIP to autonomous systems that are fewer than 15 hops in diameter. Recall that in DV protocols, neighboring routers exchange distance vectors with each other. The distance vector for any one router is the current estimate of the shortest path distances from that router to the subnets in the AS. In RIP, routing updates are exchanged between neighbors approximately every 30 seconds using a **RIP response message**. The response message sent by a router or host contains a list of up to 25 destination subnets within the AS, as well as the sender's distance to each of those subnets. Response messages are also known as **RIP advertisements**.

Let's take a look at a simple example of how RIP advertisements work. Consider the portion of an AS shown in Figure 4.35. In this figure, lines connecting the routers denote subnets. Only selected routers (*A*, *B*, *C*, and *D*) and subnets (*w*, *x*, *y*,